Advanced Computer Networks – Exam-Oriented Study Notes

Connection-Oriented vs Connectionless Services

Connection-oriented services establish a dedicated path before data transfer, ensuring reliable and ordered delivery. Connectionless services send packets independently without prior setup, offering faster transmission but no delivery guarantees. TCP is the most common connection-oriented protocol, while UDP is connectionless.

TCP – Transmission Control Protocol

TCP ensures reliable communication using sequencing, acknowledgments, and retransmissions. It uses a three-way handshake (SYN, SYN-ACK, ACK) to establish connections and flow control to avoid congestion. TCP is widely used for web browsing, email, and file transfer.

Data Link Control

The data link layer ensures reliable data transfer using flow control and error control mechanisms. Flow control prevents buffer overflow, while error control detects and retransmits corrupted frames using ARQ protocols.

ARQ Protocols

Stop-and-Wait ARQ sends one frame at a time and waits for acknowledgment, making it simple but inefficient. Go-Back-N ARQ allows multiple frames but retransmits all frames after an error. Selective Repeat ARQ retransmits only erroneous frames, improving efficiency.

Network Delays

Packet delay occurs due to processing, queuing, transmission, and propagation delays. Congestion increases queuing delay, while long distances increase propagation delay. Understanding delays is crucial for performance analysis.

Internet Protocol (IP)

IP provides logical addressing and routing at the network layer. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses to overcome address exhaustion. IP packets contain headers with routing and control information.

IPv6 Overview

IPv6 was developed to solve IPv4 address exhaustion and improve routing efficiency. It supports larger address space, simplified headers, built-in security, and auto-configuration. IPv6 eliminates broadcast and relies on multicast communication.

WAN Technologies

Wide Area Networks connect geographically distant networks using carrier services. Common WAN technologies include leased lines, circuit switching, packet switching, and virtual circuits. WANs operate mainly at the physical, data link, and network layers.

Optical Transmission and Multiplexing

Optical transmission uses light signals over fiber optic cables to achieve high-speed communication. Multiplexing combines multiple signals into one channel using techniques such as FDM, TDM, and WDM. WDM is widely used in optical networks.

NAT, PAT, and DHCP

Network Address Translation (NAT) maps private IP addresses to public IPs. PAT allows multiple devices to share a single public IP using port numbers. DHCP automatically assigns IP addresses, reducing manual configuration.

Spanning Tree Protocol (STP)

STP prevents Layer 2 loops by disabling redundant paths in switched networks. It selects a root bridge based on the lowest bridge ID and determines root and designated ports. STP ensures loop-free and stable network topology.

Quick Revision Summary

• TCP = reliable, connection-oriented communication • UDP = fast, connectionless communication • ARQ protocols ensure error-free data delivery • IPv6 solves IPv4 limitations • WAN connects distant networks • STP prevents network loops • NAT/PAT conserve IP addresses